Marmara University Engineering Faculty Electrical and Electronic Engineering Department EE 4046 Electrical Distribution Systems Mid-term

Name : Surname: Student Id:

Duration: 60 minutes 07/04/2026

Closed Book/Closed Notes. A simple calculator is allowed. No other electronics and no cell phones are permissible

Q1	Q2	Q3	Q4	Q5
40	30	10	10	10

Q.1.) A load draws 100kW with a leading pf of 0.85. What are ø (power factor angle), Q and $|S|$? Draw the power triangle

Q.2.) There are three models based on the length of transmission lines

Q.3.) Write the following voltage levels for Türkiye.

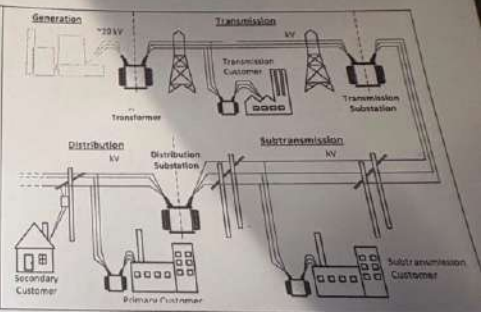
Generation ……..,

transmission ……..

Sub transmission ….,

distribution ….. ,

and end-user …



Q.4.) What is the main goal of distribution system planning?

Q.5.) Most common distribution system structure is …… why?